

Motor Datasheet: WxF 115x30HP mk2

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	113.0	mm
Machine Inner Diameter	75.7	mm
Machine Thickness / Width	24.2	mm
Air Gap	0.50	mm
Fill Factor (input)	0.60	-
Slot Fill Factor (calculated)	Not calculated	-
Slot Width (tooth to tooth)	4.416	mm
Available Coil Space	2.208	mm
Assumed Coil Width	2.008	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	757.4	g
Rotor Mass	166.7	g
Stator Mass	590.6	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	3811.0	g·cm ²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	44.632	A

Supply Current (Peak)	63.120	A
Id (Peak)	-0.000	A
Iq (Peak)	63.120	A
Phase Advance Angle	0.0	°elec
Phase Resistance	17.554	mOhm
D-Axis Inductance (Ld)	56.936	μH
Q-Axis Inductance (Lq)	55.206	μH
Terminal Inductance (Ltt = Ld + Lq)	112.141	μH
PM Flux Linkage	4.649	mWb
Frequency at operating point (electrical)	791.667	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V

Performance

Parameter	Value	Unit
Torque (Continuous)	9.44	Nm
Speed	2500.0	rpm
BackEMF Constant (Ke, phase peak)	9.2	mV/rpm
Motor Constant (Km, only resistive losses)	921.9	mNm/√W
Specific Motor Constant (Km / mass)	1.22	Nm/√W/kg
Torque Constant (Kt, peak, no-load)	133.5	mNm/A
Torque Ripple	3.1	%
Saturation	0.0	%

Losses

Parameter	Value	Unit
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Total Losses	202.0	W
Resistive Losses	104.9	W
Iron Core Losses	54.8	W
Magnet Eddy Current Losses	11.3	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	36	-
Pole Pairs	19	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	6.0	-
Number of Phase Turns (all phases in series)	72.0	-

Materials

Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-

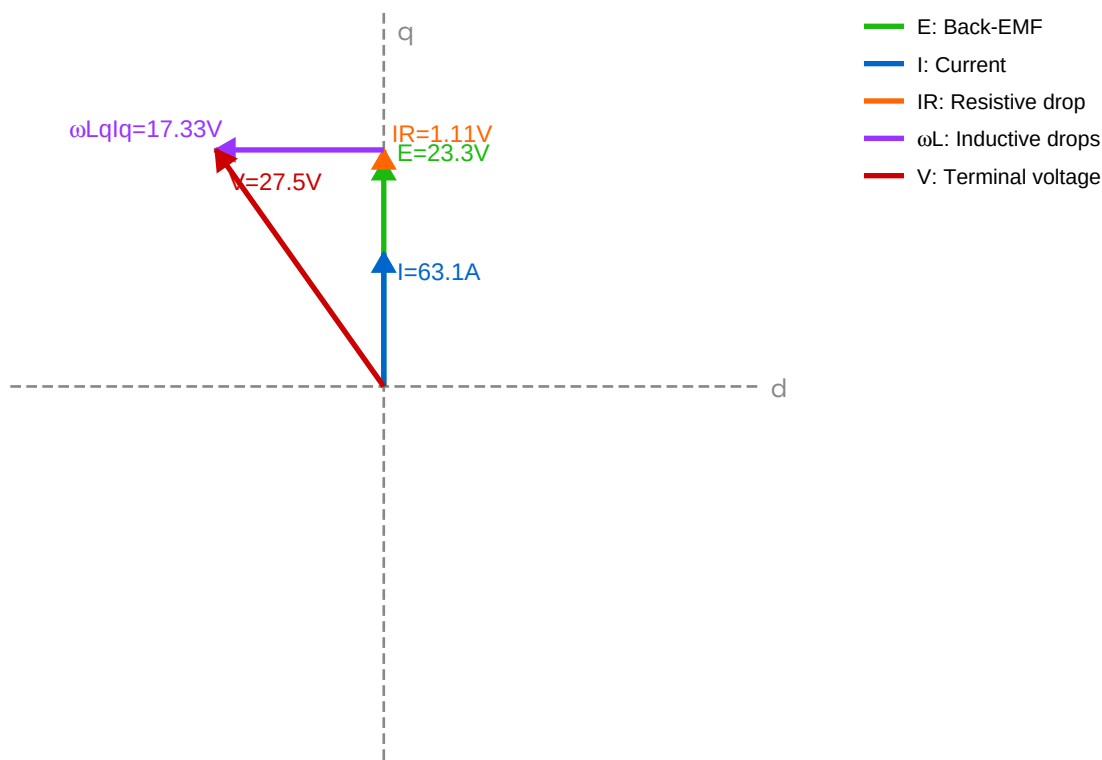
Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	23.305	V
Resistive Drop (Phase, Peak)	1.108	V

Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	17.333	V
Inductive Drop Total (Peak)	17.333	V
Terminal Voltage (Phase, Peak)	27.536	V

Back-EMF Voltage (Line-to-Line, Peak)	40.365	V
Resistive Drop (Line-to-Line, Peak)	1.919	V
Inductive Drop (Line-to-Line, Peak)	30.021	V
Terminal Voltage (Line-to-Line, Peak)	47.694	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



3D Model

